

Introduction to Dynamic Web Content

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www.wa4e.com





Browser

HTML
CSS
DOM
JavaScript
jQuery



Web Server

PHP
MySQL
Apache

<http://www.dr-chuck.com/page1.htm>

HTTP - Hypertext Transfer Protocol

- The dominant Application Layer Protocol on the Internet
- Invented for the Web - to retrieve HTML, Images, Documents, etc.
- Extended to handle data in addition to documents - RSS, Web Services, etc.
- Basic Concept: Make a connection - Request a document - Retrieve the document - Close the connection

<http://en.wikipedia.org/wiki/Http>

Uniform Resource Locator

http://	www.dr-chuck.com/	page1.htm
protocol	host	document

Getting Data from the Server

- Each time the user clicks on an anchor tag with an href = value to switch to a new page, the browser makes a connection to the web server and issues a “GET” request - to GET the content of the page at the specified URL.
- The server returns the HTML document to the browser, which formats and displays the document to the user.

Web Server

80



Browser



Web Server

80



Browser



Click

Request

Web Server

80

GET http://www.dr-chuck.com/page2.htm



Browser



Click

Request

Web Server

Response

GET http://www.dr-chuck.com/page2.htm

80

<h1>The Second
Page</h1> <p>If you like,
you can switch back to the
First
Page.</p>

Browser



Click

Request

Web Server

Response

GET http://www.dr-chuck.com/page2.htm

80

<h1>The Second
Page</h1><p>If you like,
you can switch back to the
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Page.</p>

Browser

Click

Parse/
Render



Internet Standards

- The standards for all of the Internet protocols (inner workings) are developed by an organization
- Internet Engineering Task Force (IETF)
- www.ietf.org
- Standards are called “RFCs” - “Request for Comments”

INTERNET PROTOCOL

DARPA INTERNET PROGRAM

PROTOCOL SPECIFICATION

September 1981

The internet protocol treats each internet datagram as an independent entity unrelated to any other internet datagram. There are no connections or logical circuits (virtual or otherwise).

The internet protocol uses four key mechanisms in providing its service: Type of Service, Time to Live, Options, and Header Checksum.

Source:

<http://tools.ietf.org/html/rfc791>

<http://www.w3.org/Protocols/rfc2616/rfc2616.txt>

Network Working Group
Request for Comments: 2616
Obsoletes: 2068
Category: Standards Track

R. Fielding
UC Irvine
J. Gettys
Compaq/W3C
J. Mogul
Compaq
H. Frystyk
W3C/MIT
L. Masinter
Xerox
P. Leach
Microsoft
T. Berners-Lee
W3C/MIT
June 1999

Hypertext Transfer Protocol -- HTTP/1.1

Status of this Memo

This document specifies an Internet standards track protocol for the Internet community, and requests discussion and suggestions for improvements. Please refer to the current edition of the "Internet Official Protocol Standards" (STD 1) for the standardization state and status of this protocol. Distribution of this memo is unlimited.

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Abstract

The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information

5 Request

A request message from a client to a server includes, within the first line of that message, the method to be applied to the resource, the identifier of the resource, and the protocol version in use.

```
Request      = Request-Line           ; Section 5.1
               *(( ( general-header    ; Section 4.5
                   | request-header     ; Section 5.3
                   | entity-header ) CRLF) ; Section 7.1
               CRLF
               [ message-body ]        ; Section 4.3
```

5.1 Request-Line

The Request-Line begins with a method token, followed by the Request-URI and the protocol version, and ending with CRLF. The elements are separated by SP characters. No CR or LF is allowed except in the final CRLF sequence.

```
Request-Line  = Method SP Request-URI SP HTTP-Version CRLF
```

Making an HTTP Request

- Connect to the server like www.dr-chuck.com
 - a "handshake"
- Request a document (or the default document)
 - GET <http://www.dr-chuck.com/page1.htm> HTTP/1.0
 - GET <http://www.mlive.com/ann-arbor/> HTTP/1.0
 - GET <http://www.facebook.com> HTTP/1.0

```
$ telnet data.pr4e.org 80
```

```
Trying 74.208.28.177...
```

```
Connected to data.pr4e.org character is '^]'.
```

```
GET http://data.pr4e.org/page1.htm HTTP/1.0
```

```
HTTP/1.1 200 OK
```

```
Date: Thu, 04 Jan 2018 14:45:10 GMT
```

```
Server: Apache/2.4.7 (Ubuntu)
```

```
Last-Modified: Mon, 15 May 2017 11:11:47 GMT
```

```
Content-Type: text/html
```

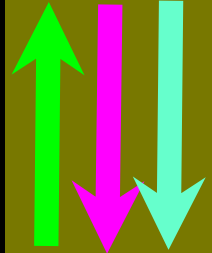
```
<h1>The First Page</h1>
```

```
<p>If you like, you can switch to
```

```
the <a href="http://www.dr-chuck.com/page2.htm">Second  
Page</a>.</p>
```

```
Connection closed by foreign host.
```

Web Server



Browser

Accurate Hacking in the Movies

- Matrix Reloaded
- Bourne Ultimatum
- Die Hard 4
- ...

<http://nmap.org/movies.html>



```
80/tcp    open      http
81/tcp    open      https
10.0.0.1   [mobile]
11 # nmap -u -ss -O 10.2.2.2
11
12 Starting nmap U. 2.54BETA2S
13 Insufficient responses for TCP sequencing (3). OS detection i
13 accurate
14 Interesting ports on 10.2.2.2:
44 (The 1539 ports scanned but not shown below are in state: cl
51 Port      State      Service
51 22/tcp    open      ssh
58
68 No exact OS matches for host
68
74 Nmap run completed -- 1 IP address (1 host up) scanned
50 # ssh -u 10.2.2.2 -rootpw="210H0101"
Connecting to 10.2.2.2:ssh ... successful.
Re Attempting to exploit sshv1 CRC32 ... successful.
IP Resetting root password to "210H0101".
System open: Access level (9)
No # ssh 10.2.2.2 -l root
root@10.2.2.2's password:
[REDACTED]
ACCESS GRANTED
```

Getting to Know Our Browsers

Find Developer Mode

- Chrome: View > Developer
- FireFox: Tools -> Web Developer -> Toggle
- Safari: Preferences > Advanced > Show Develop Menu

Dr. Chuck's Awesome H...
www.dr-chuck.com

Apps H Tw CT CV CSites CSera LTI Forum IS IEEE Liberal Arts and Sci Blue Other Bookmarks

New: Tsugi: A PHP framework for IMS LTI Tools
New: MOOCs: Charles Severance at TEDxKalamazoo

Free Courses / Educational Material:
Coursera: Programming for Everybody (Python)
Coursera: Internet History, Technology and Security
SI 502 - Networked Computing
See also www.pythonlearn.com, www.php-intro.com and www.appenginelearn.com

Books
Python For Informatics: Exploring Information (2010, 2014)
Sakai: Building an Open Source Community (2011, 2014)
Raspberry Pi (21st Century Skills Innovation Library) (2013)
Using Google App Engine (O'Reilly 2009)
High Performance Computing (O'Reilly 1998, Connexions 2010)

Web/Multimedia sites

Tweets Follow
Jason Cidras @JasonCidras
@drchuck I just signed up for your Python class, I am very excited to finally learn it! Thanks for the opportunity!
Retweeted by drchuck
Expand
drchuck @drchuck
Just a reminder...

School of Information
Rate this Professor

Blog
@drchuck Twitter
Keynote Speaker

Elements Network Sources Timeline Profiles Resources Audits Console

Preserve log Disable cache

Name Path	Method	Status Text	Type	Initiator	Size Content	Time Latency	Timeline
www.dr-chuck.com	GET	200 OK	text/...	Other	10.8 KB 10.6 KB	173 ms 172 ms	
csev_ian_dolphin_small.jpg	GET	200 OK	imag...	www.dr-ch... Parser	18.3 KB 18.1 KB	115 ms 58 ms	
rate-my-professor.jpg /images	GET	200 OK	imag...	www.dr-ch... Parser	11.8 KB 11.5 KB	114 ms 113 ms	
icft=drchu02-2016mar2&no=1		200		www.dr-ch	159 B	31 ms	

58 requests | 697 KB transferred | 17.14 s (load: 1.36 s, DOMContentLoaded: 585 ms)

Console Search Emulation Rendering

01eviz.zip Show All

DEMO

Request

GET http://www.dr-chuck.com/page2.htm

Web Server

80



Response

```
<h1>The Second  
Page</h1><p>If you like,  
you can switch back to the  
<a href="page1.htm">First  
Page  
</a>.</p>
```

Browser



Click

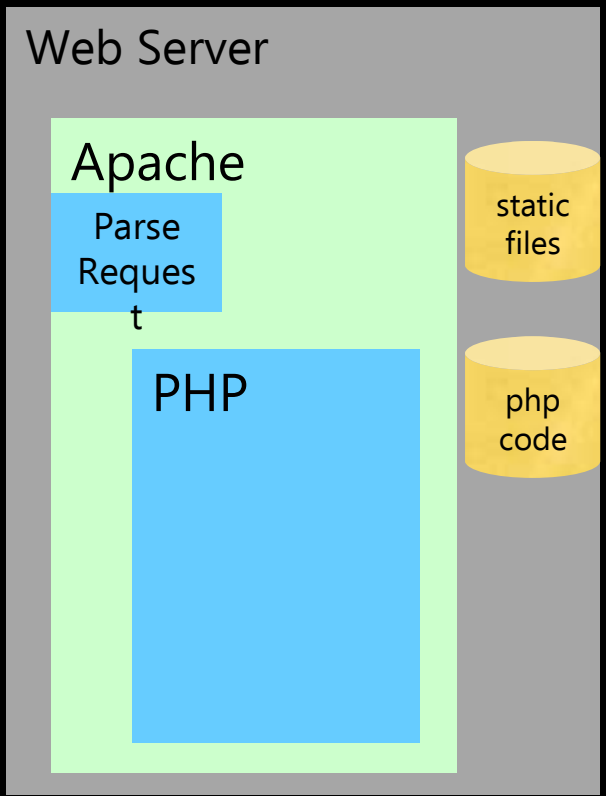
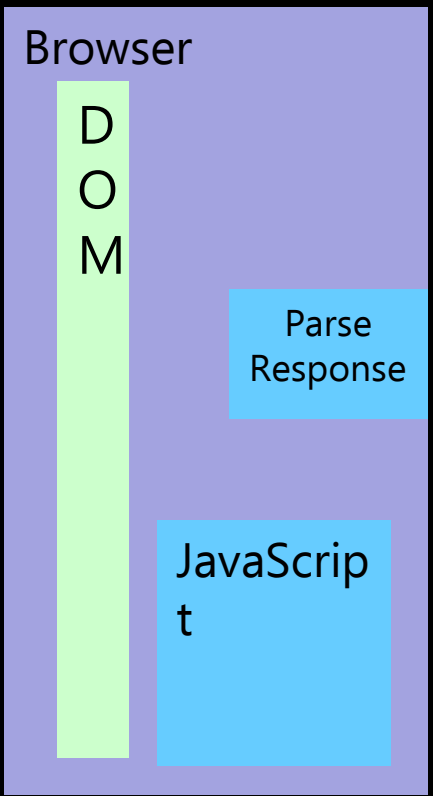
Parse/
Render



This is Only the Beginning...

- In the Browser – HTML, CSS, Document Object Model (the DOM) and JavaScript
- In the Web Server – Apache (or similar) web server with a PHP extension installed
- In the Database Server running MySQL - PostgreSQL, Oracle, SQL Server, or similar

Time



RRC/HTTP

SQL

Summary

- Many modern web applications use a stack of open source software working together.
- While you can use frameworks to accelerate productivity, this course focuses on what is really going on so you can fully understand what frameworks are doing on your behalf.

Acknowledgements / Contributions



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Continue new Contributors and Translators here

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Insert new Contributors and Translators here including names and dates